Dual-Branch CNN With Enhanced Bidirectional Multiscale Sparse Cross-Attention for Remote Sensing Scene Classification

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***Abstract*—Remote sensing scene classification (RSSC) remains challenging due to large intraclass diversity and high interclass similarity, while a single backbone often struggles to capture both global semantics and local structures. This article proposes a lightweight dual-branch convolutional neural network (CNN) equipped with an enhanced bidirectional multiscale sparse cross-attention (EBiMSC) module for multiscale feature fusion. The proposed network integrates two complementary branches: a streamlined ConvNeXt-Tiny branch that enhances global contextual modeling and long-range dependency capture, and a lightweight GhostNetV3 branch that captures fine-grained textures and local details, thereby reducing confusion among visually similar categories. To strengthen cross-branch complementarity, the EBiMSC module performs bidirectional multiscale sparse cross-attention (MSC) to facilitate interaction across different scales, while a parallel lightweight enhancement path supplements informative local cues. The fused features are adaptively integrated through learnable weights to obtain discriminative and consistent representations, followed by an attention-guided lightweight multilayer perceptron (MLP) head for classification. Despite its concise design, the model is computationally efficient, with 19.5 million parameters and 4.24 GFLOPs, making it suitable for real-world applications. Extensive experiments on four benchmark RSSC datasets demonstrate that the proposed method achieves superior performance compared with recent CNN-based, Transformer-based, and hybrid models, validating its effectiveness and efficiency.**

***Index Terms*—Dual-branch convolutional neural network (CNN), lightweight networks, multiscale feature fusion, remote sensing scene classification (RSSC), sparse cross-attention.**

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I. Introduction

REMOTE sensing scene classification (RSSC) aims to assign category labels to remote sensing (RS) images by semantically analyzing scene content, including surface textures, spatial layouts, and object relationships. As an important task in RS image analysis, RSSC underpins many practical applications such as land-use change monitoring, environmental assessment and protection, disaster response, and urban planning.

With the continuous advancement of deep learning, convolutional neural networks (CNNs) [1] have achieved remarkable success in various computer vision tasks, such as image classification, object detection, and semantic segmentation, and have been widely applied to RSSC. Traditional CNN architectures, such as VGGNet [2] and ResNet [3], have demonstrated strong performance in RSSC tasks. However, RS scenes typically exhibit high interclass similarity and large intraclass diversity, posing inherent challenges to discriminative representation learning. On the one hand, different scene categories may exhibit highly similar local textures and spatial layouts, as shown in Fig. 1, where farmland, forest, and meadow often contain extensive green regions with comparable structural patterns, thereby blurring semantic distinctions. As a result, samples from different classes tend to overlap in the feature space, weakening interclass boundaries. On the other hand, the same category can exhibit substantial intraclass variation, as shown in Fig. 2, due to changes in imaging scale, geographic context, seasonal illumination, and object distribution. Taking sparse residential scenes as an example, variations in building density, roof materials, and surrounding vegetation across regions further hinder models from learning stable and generalizable features.

Meanwhile, various attention mechanisms have been introduced to enhance feature extraction and contextual representation in RSSC. Representative examples include channel attention [4], [5], spatial attention [6], and hybrid modules such as convolutional block attention module (CBAM) [7], triplet attention [8], and group CBAM enhancer (GCE) [3]. Incorporating these modules improves a model's ability to emphasize informative regions in RS images. However, RS scenes exhibit richer semantics, larger scale variations, and more diverse object morphologies than natural images. Consequently, attention plugged into a single backbone offers

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Fig. 1. Examples of interclass similarity in RS scenes. Different land-cover categories such as (a) farmland, (b) forest, and (c) meadow share visually similar green textures and spatial layouts, which can easily cause category confusion.

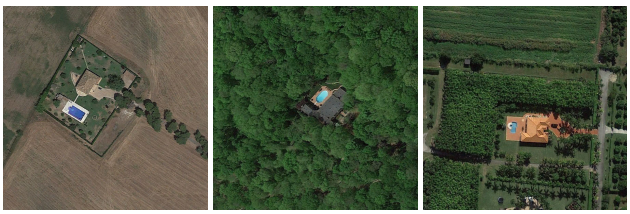


Fig. 2. Examples of intraclass diversity in RS scenes. Different sparse residential scenes such as (a) open-field sparse residential, (b) woodland-surrounded sparse residential, and (c) sparse residential with regularly patterned surroundings exhibit visible variations in roof appearance, surrounding vegetation, and spatial layout.

only limited cross-scale interaction and often fails to reconcile global semantics with fine-grained details in RSSC.

In recent years, RSSC performance has been advanced by CNN–Transformer hybrids, multibranch fusion, and semantic- or distillation-guided training. Despite their effectiveness, these methods typically rely on heavier backbones or extra training pipelines, raising computational cost and hindering practical deployment. Therefore, rather than further escalating model complexity, we revisit the design principles of compact convolutional architectures to explore a CNN-based solution that explicitly decouples global semantics and local details and then recouples them through an efficient multiscale cross-branch interaction that suppresses redundant responses under tight computational budgets.

Motivated by these challenges, this study investigates three key questions.

1) How can global semantics and fine-grained details be jointly modeled within a compact CNN framework?
2) How can a lightweight CNN-based architecture achieve strong RSSC performance without auxiliary supervision or distillation?
3) How can an efficient multiscale fusion mechanism be devised to enhance complementarity between global semantic and detail-sensitive representations while suppressing redundant interactions?

To address these issues, we propose a dual-branch CNN with enhanced bidirectional multiscale sparse cross-attention (DBEBiMSC) model for RSSC, with convolutional backbones where ConvNeXt-Tiny captures global semantics, and GhostNetV3 preserves fine-grained details. An EBiMSC module performs efficient bidirectional sparse multiscale fusion between branches, and the fused features are classified by an attention-guided lightweight multilayer perceptron (MLP) head.

The main contributions of this article are summarized as follows.

1) We propose DBEBiMSC, a lightweight dual-branch CNN with two complementary convolutional backbones for RSSC. By jointly modeling global semantics and fine-grained local details within a compact architecture, DBEBiMSC achieves a favorable accuracy–efficiency tradeoff compared with recent Transformer-based and hybrid methods.
2) We construct a heterogeneous complementary backbone coupling ConvNeXt-Tiny and GhostNetV3, where ConvNeXt captures high-level semantics and GhostNetV3 preserves fine-grained spatial details. Triplet attention is employed to strengthen cross-dimensional interactions and stabilize branch complementarity.
3) We develop BiMSC and EBiMSC for achieving heterogeneous dual-branch fusion. Specifically, BiMSC reformulates sparse multiscale fusion from one-way intrabackbone refinement into bidirectional cross-branch interaction, enabling explicit mutual calibration between semantic and detail streams. Moreover, EBiMSC further introduces a parallel enhanced fusion path with adaptive weighting for more robust semantic–detail integration.
4) Extensive experiments on four benchmark datasets under multiple training ratios demonstrate that DBEBiMSC matches or surpasses state-of-the-art (SOTA) CNN, Transformer, and hybrid models while maintaining a substantially more compact model size and lower FLOPs.

II. Related Work

A. Remote Sensing Scene Classification

Early RSSC research relied on handcrafted descriptors such as color histograms (CHs) [9], scale-invariant feature transform (SIFT) [10], histograms of oriented gradients (HOGs) [11], and bag-of-visual-words (BoVW) [12] representations. These approaches mainly capture low- and mid-level cues, leading to limited discriminative power and weak adaptability to complex real-world scenes.

With the advent of deep learning, neural network-based RSSC methods have achieved substantially improved performance by learning high-level representations from RS images. Among them, CNNs have been the most widely adopted, progressively aggregating local cues into global semantics through hierarchical convolutions and thus providing strong spatial invariance and semantic expressiveness. Classical CNN backbones such as VGG [2], GoogLeNet [13], and ResNet [3] have reported strong performance on multiple RS benchmarks. More recently, efficient CNN backbones originally developed for mobile/edge vision, such as MobileOne [14] and StarNet [15], have been explored as lightweight baselines for RSSC, further highlighting the importance of accuracy–efficiency tradeoffs. However, recent RSSC benchmarks suggest that these ultralight designs may not consistently transfer well to RS scenes. Because aggressive compression can discard

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high-frequency details and weak textures, and simplified feature-mixing operations provide limited explicit long-range context modeling, which makes it harder to distinguish categories with complex layouts and small objects [16], [17].

Nevertheless, conventional single-backbone CNNs are limited in modeling long-range dependencies and global context. The inherent locality of convolutions makes it difficult to jointly preserve global semantic coherence and fine-grained spatial details, which are crucial for RSSC under multiscale targets and high intraclass diversity. In recent years, Transformers [18] have been introduced into RS thanks to their strong global modeling ability and capacity for long-range dependency capture. Models such as ViT [19], Swin Transformer [20], CSwin Transformer [21], and SCViT [22] exhibit strong semantic modeling on large-scale datasets, alleviating CNNs' limited receptive fields. However, they are often data- and compute-intensive and may underperform in capturing local textures and fine-grained structures. Therefore, recent RSSC studies have explored hybrid fusion or enhanced CNN paradigms to balance global modeling and local detail, yet a persistent tradeoff between accuracy and deployment efficiency remains.

B. CNN-Based Methods for RSSC

In recent years, RSSC has made substantial progress driven by deep learning. CNNs have remained central to RSSC due to their strong hierarchical feature abstraction. As model scales and application demands grow, CNN architectures have evolved from traditional designs toward more efficient modern backbones. MobileNet [23] employs depthwise separable convolutions for lightweight design, EfficientNet [24] introduces compound scaling to balance depth, width, and resolution, and GhostNet [25] adopts ghost modules for cheap feature generation with fewer parameters. ConvNeXt [26] adopts Transformer-inspired large kernels and LayerNorm, strengthening CNNs' global context modeling. Such backbones are well suited for constructing heterogeneous multibranch designs that explicitly model global–local complementarity.

To cope with complex layouts, scale variations, and background interference in RSSC, numerous enhanced CNN-based models have been developed to strengthen spatial–semantic representation. Liu et al. [2] proposed a global latent relationship-embedded CNN (GLR-CNN) built on VGG16, which accepts arbitrary-size HRSIs and refines scene semantics via attention-based multilayer feature fusion. Zhao et al. [27] proposed the co-enhanced global–part integration network (CGINet), which introduces a light context-aware attention block for contextual interaction and a co-enhanced loss module for adaptive supervision, improving feature discrimination for RSSC. Zhang et al. [28] proposed a distributed convolutional neural network (DCNNet) for edge–cloud collaborative RSSC, featuring progressive early exit inference with distributed self-distillation, and integrating multiscale feature fusion and sampling augmentation attention (SA) to enhance robustness. Ma et al. [3] proposed the multiscale sparse cross-attention network (MSCN), which embeds a multiscale sparse cross-attention (MSC) module and a GCE into a single ResNet50 backbone to fuse multilevel features and enhance discrimination in a lightweight manner.

Despite these advances, CNN-based RSSC methods still struggle to jointly model global semantics and fine-grained local structures under lightweight constraints. While modern backbones improve efficiency and enhanced CNN frameworks strengthen feature discrimination, most existing designs still operate within a single backbone. In particular, multiscale fusion schemes such as MSC are mainly formulated as one-way intrabackbone refinement, making them less suitable for explicit interaction between heterogeneous semantic and detail streams. This limitation motivates our BiMSC/EBiMSC for bidirectional cross-branch fusion in a compact dual-branch CNN.

C. Multibranch Fusion and Cross-Scale Modeling for RSSC

To better capture global semantics and local details beyond single-backbone, recent studies have explored multibranch and cross-scale fusion architectures to enhance heterogeneous feature complementarity and semantic coherence. Zheng et al. [29] proposed a lightweight dual-branch Swin Transformer (LDBST), which splits stage-wise features into a Swin Transformer branch (with Conv-MLP enhancement) and a lightweight CNN branch to jointly model global context and local details at low cost. Yang et al. [30] proposed the local–global lifting Transformer (LGLFormer), which combines a multiscale CNN branch and a Transformer-based global branch guided by a 2-D local–global adaptive feature extractor to model long-range local–global dependencies. Yang et al. [31] proposed the spatial–frequency multiscale Transformer (SF-MSFormer), a framework that enhances global feature representation across multiple scales. Xie et al. [32] proposed ACTFormer, an attention–convolution synergy Transformer that progressively stacks local and global ACT blocks with Conv-MLP to integrate fine-grained details and global context for RSSC. Chai et al. [33] proposed the label semantic dynamic guidance network (LSDGNet), which introduces large language model (LLM)-derived label semantics via an auxiliary cross-modal guidance branch and employs a dynamic soft label module and a coarse-to-fine contrastive module to enhance discrimination for confusing scenes. Dong et al. [16] proposed LMHMamba, a lightweight multifeature hybrid model whose core LMHM module adopts a four-branch design to integrate StarNet-style elementwise multiplication, local attention, HSM-SSD state-space modeling, and global attention for joint modeling of local details and long-range dependencies in RSSC.

Beyond explicit multibranch designs, some studies instead pursue cross-layer and cross-scale fusion *within a single backbone* to improve cross-level semantic alignment and discriminability. Tang et al. [34] proposed efficient multiscale Transformer and cross-level attention learning (EMTCAL), which couples an efficient multiscale Transformer with a cross-level attention module to aggregate multilevel features for semantic interaction across layers. Zhang et al. [35] proposed the multiple hierarchical cross-scale Transformer (MHCT), which models multilevel cross-scale dependencies

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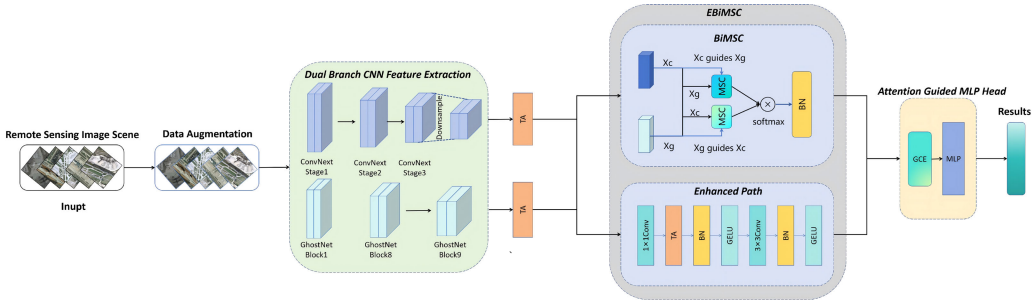


Fig. 3. Framework of the proposed DBEBiMSC network. A dual-branch backbone employs ConvNeXt-Tiny to capture global semantics and GhostNetV3 to extract local details, with triplet attention applied to both branches to enhance cross-dimensional interactions. The EBiMSC module performs multiscale cross-branch fusion with an adaptive weighting mechanism, and the fused features are further refined by GCE and then classified by a lightweight MLP head.

via hierarchical attention to mitigate scale-induced feature misalignment. Duan et al. [36] proposed a Swin Transformer with multiscale fusion (STMSF), which exploits the hierarchical Swin backbone and a SAPN-enhanced fusion module to jointly strengthen global context modeling and fine-grained multiscale detail representation for RSSC.

Multibranch and cross-scale fusion has advanced RSSC by improving global–local complementarity and cross-scale semantic interaction. However, limitations remain in lightweight settings, as recent methods often rely on Transformer-based hybrids, auxiliary branches, or more complex fusion pipelines, while many existing fusion schemes remain one-way or single path. In contrast, EBiMSC reformulates sparse multiscale fusion as bidirectional cross-branch calibration between semantic and detail streams, and further complements the sparse attention route with a parallel enhanced fusion path without Top-k pruning. With adaptive weighting between the sparse and enhanced routes, this design provides a more robust lightweight solution for semantic–detail integration in RSSC.

III. Method

A. Overall Framework

To overcome the inability of single-backbone networks to capture global semantics and fine-grained details simultaneously, we propose a dual-branch network termed DBEBiMSC. As shown in Fig. 3, the model employs two heterogeneous backbones and an EBiMSC module for cross-branch interaction and multilevel complementary fusion.

DBEBiMSC comprises two parallel branches: a ConvNeXt-Tiny semantic branch and a lightweight GhostNetV3 detail branch. The ConvNeXt-Tiny branch is responsible for modeling high-level semantics and global contextual relationships of the scene, while the GhostNetV3 branch focuses on efficient extraction of fine-grained structures such as edges, textures, and small objects. Triplet attention is inserted into both branches to enhance spatial and channel-wise representations. Their outputs are projected to a unified embedding dimension via 1×1 convolutions for channel alignment, yielding a compact joint feature space for fusion. EBiMSC is then applied for fusion. Built on BiMSC, EBiMSC adds an enhanced fusion path and learnable adaptive weights to integrate both branches more effectively, enabling discriminative multiscale aggregation from global and local perspectives. After fusion, GCE emphasizes salient responses and suppresses redundancy. Finally, the refined features are passed through normalization layers and a lightweight MLP head for classification.

Overall, DBEBiMSC couples heterogeneous backbones via adaptive bidirectional multiscale fusion and a lightweight attention-enhanced classifier for efficient RSSC.

B. Dual-Branch Backbone Network

Following the overall design in Section III-A, we instantiate a dual-branch backbone with ConvNeXt-Tiny and GhostNetV3 as a compact yet complementary feature extractor.

The first branch employs ConvNeXt-Tiny to model high-level semantics and global context of RS scenes. In practice, we retain the early ConvNeXt-Tiny layers (the first three stages and the downsampling layer of stage four), producing a compact feature map with a large receptive field while preserving discriminative spatial structures. Initialized with ImageNet-1K [37] pretrained weights, this branch serves as the semantic backbone, focusing on large-scale layouts and category-level context.

The second branch introduces GhostNetV3 to provide lightweight yet detail-sensitive representations. We reuse the GhostNetV3 convolutional extractor, discarding global pooling and the classification head. This branch yields detail-sensitive representations at low computational cost. It captures edges, textures, shapes, and other fine-grained patterns, complementing ConvNeXt in modeling small-scale targets and subtle appearance variations. Its parameters are also initialized from ImageNet-1K for stable optimization.

Triplet attention is appended to both branch outputs to capture spatial and channel-wise dependencies. Each branch output is projected to 128 channels via a 1×1 convolution, forming a unified embedding space for subsequent EBiMSC fusion.

Overall, the dual-branch backbone combines ConvNeXt-Tiny for semantic modeling and GhostNetV3 for efficient detail extraction, yielding complementary multiscale representations under a lightweight purely convolutional framework.

C. Enhanced Bidirectional Multiscale Sparse Cross-Attention

In dual-branch architectures, fusing semantic and detail features effectively remains a significant challenge due to their heterogeneous characteristics and the risk of asymmetric

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information flow. To address this challenge, we propose the EBiMSC module to improve the representational capacity and discriminative power of dual-branch feature fusion. Our ablation studies show that directly inserting the original single-direction MSC into the dual-backbone architecture yields only marginal gains and may even degrade performance, indicating that a one-way sparse cross-attention pipeline is prone to branch dominance and cannot reliably exploit complementary cues when the two streams encode different inductive biases, with one emphasizing global semantics and the other focusing on fine-grained details. To overcome this limitation, EBiMSC introduces a dual-path bidirectional fusion design with three key components: 1) a *bidirectional and structurally mirrored* MSC scheme with independent parameters, which enables mutual guidance between the two branches and stabilizes cross-branch interaction; 2) a *parallel enhanced fusion path* that aggregates features from both branches and performs channel compression, attention-based refinement, and local convolutional enhancement to compensate for information loss caused by sparsification in the main attention route; and 3) a *hierarchical adaptive weighting mechanism* that balances the contributions of different sparsity paths within each directional stream, the two opposite interaction directions in BiMSC, and the two parallel fusion routes, namely the main attention path and the enhanced path.

1) BiMSC: Considering the structural diversity of cross-scale targets in RS scenes, BiMSC extends MSC with a bidirectional interaction strategy between two heterogeneous backbones. Let $\mathbf{X}_c$ and $\mathbf{X}_g$ denote the feature maps from the ConvNeXt and GhostNet branches, respectively. BiMSC instantiates two MSC blocks with mirrored designs to perform opposite-direction cross-attention

$$\mathbf{X}_c \to \mathbf{X}_g: \text{ queries from } \mathbf{X}_c, \text{ keys/values from } \mathbf{X}_g$$
$$\mathbf{X}_g \to \mathbf{X}_c: \text{ queries from } \mathbf{X}_g, \text{ keys/values from } \mathbf{X}_c.$$

We define $c \to g$ as the cross-attention direction with queries from $\mathbf{X}_c$ and keys/values from $\mathbf{X}_g$; $g \to c$ is defined analogously. The fused output in each direction is aligned with the query branch.

In each direction, MSC operates in a sparse multiscale manner. Concretely, we build lightweight multiscale context embeddings of the *source* features using multiscale aggregation with kernel sizes $\{3, 5, 7\}$ before computing key/value representations, enabling each query to aggregate both local and broader contextual cues. To impose sparsity, we retain only the top $\lfloor N_s/k_1 \rfloor$ and $\lfloor N_s/k_2 \rfloor$ attention responses per query over source tokens, with $(k_1, k_2) = (2, 3)$ on all datasets so that performance gains mainly reflect the BiMSC/EBiMSC design rather than hyperparameter tuning. Since BiMSC/EBiMSC improves cross-branch interaction mainly through bidirectionality and heterogeneous cooperation, this unified setting serves as a robustness-oriented evaluation rather than a sparsity-driven search. Let $\mathbf{A}_{c\to g}^{(k_1)}$ and $\mathbf{A}_{c\to g}^{(k_2)}$ (similarly, $\mathbf{A}_{g\to c}^{(k_1)}$ and $\mathbf{A}_{g\to c}^{(k_2)}$) denote the normalized sparse attention maps after top-k filtering. We denote $\mathbf{V}_{c\to g}$ and $\mathbf{V}_{g\to c}$ as the directional value embeddings, defined by

$$\mathbf{V}_{c\to g} = \phi_v\left(\mathbf{X}_g\right), \qquad \mathbf{V}_{g\to c} = \phi_v\left(\mathbf{X}_c\right). \tag{1}$$

Coefficients α_1, α_2, β_1, and β_2 are learnable fusion weights for the two sparsity paths in each directional stream. The directional interaction outputs are defined as

$$\mathbf{Z}_{\text{MSC-}c\to g} = \alpha_1 \mathbf{A}_{c\to g}^{(k_1)} \mathbf{V}_{c\to g} + \alpha_2 \mathbf{A}_{c\to g}^{(k_2)} \mathbf{V}_{c\to g} \tag{2}$$
$$\mathbf{Z}_{\text{MSC-}g\to c} = \beta_1 \mathbf{A}_{g\to c}^{(k_1)} \mathbf{V}_{g\to c} + \beta_2 \mathbf{A}_{g\to c}^{(k_2)} \mathbf{V}_{g\to c}. \tag{3}$$

Finally, the bidirectional fusion output of the BiMSC module is formulated as

$$\mathbf{Z}_{\text{BiMSC}} = \gamma_1 \mathbf{Z}_{\text{MSC-}c\to g} + \gamma_2 \mathbf{Z}_{\text{MSC-}g\to c}$$
$$\left[\gamma_1, \gamma_2\right] = \text{softmax}\left(\mathbf{w}\right), \quad \gamma_1 + \gamma_2 = 1. \tag{4}$$

where $\mathbf{w} = [w_1, w_2]$ is trainable and γ_1 and γ_2 adaptively balance the two directional interaction paths.

In this work, our primary goal is to validate the bidirectional multiscale interaction architecture rather than perform dataset-specific sparsity tuning. Using a unified top-k configuration across datasets ensures that gains are attributable to BiMSC/EBiMSC design instead of hyperparameter search.

2) Enhanced Fusion Path: To further exploit discriminative local details, we introduce an enhanced fusion path in parallel with the BiMSC main route. This path concatenates $\mathbf{X}_c$ and $\mathbf{X}_g$ along channels and then applies a 1×1 convolution for compression to a unified feature space

$$\mathbf{F}_{\text{cat}} = \left[\mathbf{X}_c, \mathbf{X}_g\right] \in \mathbb{R}^{B\times 2d\times H\times W} \tag{5}$$
$$\mathbf{F}_{\text{cmp}} = \text{Conv}_{1\times 1}\left(\mathbf{F}_{\text{cat}}\right) \in \mathbb{R}^{B\times d\times H\times W}. \tag{6}$$

The compressed features are fed into triplet attention to model spatial–channel interactions and enhance cross-dimensional representations. The output is normalized and activated by GELU and then refined by a 3×3 convolution to strengthen edge awareness and discriminability. The final enhanced output of this path is denoted as $\mathbf{Z}_{\text{Enh}} \in \mathbb{R}^{d\times H\times W}$

$$\mathbf{F}_{\text{attn}} = \text{TA}\left(\mathbf{F}_{\text{cmp}}\right) \tag{7}$$
$$\mathbf{F}_{\text{ref1}} = \text{GELU}\left(\text{BN}\left(\mathbf{F}_{\text{attn}}\right)\right) \tag{8}$$
$$\mathbf{Z}_{\text{Enh}} = \text{GELU}\left(\text{BN}\left(\text{Conv}_{3\times 3}\left(\mathbf{F}_{\text{ref1}}\right)\right)\right). \tag{9}$$

To control the weighting between BiMSC and the enhanced path, we introduce a trainable fusion vector $\mathbf{u} = [u_1, u_2]$. After softmax normalization, it adaptively regulates the integration ratio of the two paths, producing the fused representation

$$\mathbf{Z}_{\text{fused}} = \lambda_1 \mathbf{Z}_{\text{BiMSC}} + \lambda_2 \mathbf{Z}_{\text{Enh}}$$
$$[\lambda_1, \lambda_2] = \text{softmax}\left(\mathbf{u}\right), \quad \lambda_1 + \lambda_2 = 1 \tag{10}$$

where λ_1 and λ_2 are learnable coefficients from $\mathbf{u}$ via softmax and optimized jointly, so the contributions of bidirectional interaction and local refinement are data-driven rather than heuristic. In particular, the enhanced path compensates for missing local details in the main BiMSC stream, consistent with our ablation results.

D. Global Prediction via MLP

To enhance the discriminability of fused features while preserving spatial consistency, we introduce an attention-guided MLP head. It strengthens feature representation and enables global prediction via attention enhancement followed by MLP

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processing. The head consists of three steps: attention-based enhancement, spatial flattening with LayerNorm, and global prediction by a lightweight MLP.

Given the fused map $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$ from EBiMSC, we first apply the lightweight GCE module to enhance salient regions, producing

$$\mathbf{F}_{\mathrm{GCE}} = \mathrm{GCE}(\mathbf{Z}). \tag{11}$$

We then flatten $\mathbf{F}_{\mathrm{GCE}}$ over spatial dimensions into a token sequence $\mathbf{X} \in \mathbb{R}^{(H\cdot W)\times C}$ for MLP processing

$$\mathbf{X} = \mathrm{Flatten}(\mathbf{F}_{\mathrm{GCE}}) \in \mathbb{R}^{(H\cdot W)\times C}. \tag{12}$$

LayerNorm is applied to each token to mitigate channel shifts and stabilize optimization

$$\mathbf{X} = \mathrm{LayerNorm}(\mathbf{X}). \tag{13}$$

The normalized tokens are fed into a lightweight two-layer MLP with GELU and Dropout, which can be written in a simplified form as

$$\mathbf{X}' = \mathrm{Dropout}(\mathrm{GELU}(\mathbf{X}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2). \tag{14}$$

We insert a lightweight 3×3 depthwise convolution between the two linear layers (Conv-FFN) and apply dropout after both the activation and the second linear layer. Here, $\mathbf{W}_1 \in \mathbb{R}^{C\times d}$, $\mathbf{W}_2 \in \mathbb{R}^{d\times N}$, and N is the number of scene categories. In our implementation, we set $d = 256$ and use a dropout ratio of 0.5 to reduce overfitting.

Finally, global average pooling over all tokens yields the scene-level prediction

$$\hat{\mathbf{y}} = \frac{1}{H\cdot W}\sum_{i=1}^{H\cdot W}\mathbf{X}'_i. \tag{15}$$

IV. Experiments

A. Dataset Introduction

To evaluate the effectiveness of the proposed method in RS image classification tasks, four representative benchmark datasets for RSSC are selected for experimental validation, namely WHU-RS19 [38], UCMerced Land-Use [12], Aerial Image dataset (AID) [39], and NWPU-RESISC45 [40].

1) WHU-RS19 Dataset: The WHU-RS19 dataset contains 19 scene categories, with approximately 50 images per category, totaling 1005 images. Each image has a resolution of 600×600 pixels.

2) UCMerced Land-Use Dataset (UCM): The UCM dataset contains 21 scene categories, with 100 images per category, resulting in a total of 2100 images. The dataset provides a spatial resolution of approximately 0.3 m, and each image has a size of 256×256 pixels.

3) Aerial Image Dataset: The AID dataset contains 30 scene categories, with approximately 220–420 images per category, totaling 10 000 images. The dataset's spatial resolution ranges from 0.5 to 8 m, and the image size is 600×600 pixels.

4) NWPU-RESISC45 Dataset (NWPU): The NWPU dataset contains 45 scene categories, with 700 images for each category, totaling 31 500 images. The dataset's spatial resolution ranges from 0.2 to 30 m, and the image size is 256×256 pixels. It is one of the largest RSSC datasets to date.

TABLE I

Comparison of OA and Std of Different Methods on WHU-RS19. The Best Values Are Highlighted in Bold

Method	Year	OA±STD(%)	
		20% Training	50% Training
(† : CNN-based methods)			
GoogLeNet [33]†	2015	78.41±0.42	90.57±1.11
VGGNet16 [33]†	2015	93.05±1.05	97.39±0.61
ResNet50 [33]†	2016	95.50±1.06	98.41±0.37
DenseNet121 [33]†	2017	96.29±1.01	98.42±0.48
MobileNetV4-Large [23]†	2024	89.29±1.18	94.56±1.23
(‡ : ViT-based or Hybrid architecture-based methods)			
ViT-B [33]‡	2020	94.06±0.75	97.38±0.49
VisFormer [33]‡	2021	92.23±1.66	96.87±0.58
DeiT [33]‡	2021	96.88±0.57	98.89±0.32
SF-MSFormer [31]‡	2023	96.63±0.23	97.80±0.39
LGLFormer [30]‡	2024	96.62	99.20±0.06
LSDGNet [33]‡	2025	97.38±0.69	99.29±0.29
Ours†	2025	**98.37±0.38**	**99.60±0.20**

B. Experimental Environment

All experiments are implemented in PyTorch on a workstation with an NVIDIA RTX 4090 GPU. DBEBiMSC is trained using AdamW with an initial learning rate of 1×10^{-4}, batch size 32, for 300 epochs. We use cosine annealing with a five-epoch linear warm-up, decaying the learning rate to 1×10^{-5} by epoch 200. The standard cross-entropy loss is used. For each dataset, images are randomly split into training and test sets by the specified ratios and resized to 224×224. To enhance generalization, we employ standard data augmentations such as random resized crops, horizontal/vertical flips, random rotations, color jitter, and random erasing, followed by ImageNet normalization. Each DBEBiMSC experiment is repeated five times with different splits, and we report the mean OA and Std for statistical reliability. We also visualize confusion matrices to analyze class-wise performance.

C. Comparison With SOTA Methods

To verify the effectiveness of DBEBiMSC, we conduct comparative experiments against SOTA methods on four public benchmark datasets. The compared approaches are grouped into three categories: CNN-based, ViT-based, and hybrid CNN–Transformer architectures. Detailed results on each dataset are presented next. Most competing results are literature-reported results, whereas MobileNetV4-Large (Conv-only) is reproduced in this work.

1) WHU-RS19 Dataset: We evaluate DBEBiMSC on WHU-RS19 against representative CNN-based, ViT-based, and hybrid methods. As shown in Table I, DBEBiMSC achieves 98.37% ± 0.38% and 99.60% ± 0.20% OA under the 20% and 50% training ratios, respectively, establishing SOTA performance. Compared with classical CNN baselines, our model yields markedly higher accuracy and consistently surpasses recent ViT and hybrid architectures, especially under limited training samples. Notably, DBEBiMSC outperforms LSDGNet by a clear margin at 20% training and still retains an advantage at 50%, validating the effectiveness of the proposed bidirectional cross-branch fusion. Fig. 4(a) and (b) shows the confusion matrices under the 20% and 50% training ratios. With only 20% training data, predictions are already strongly

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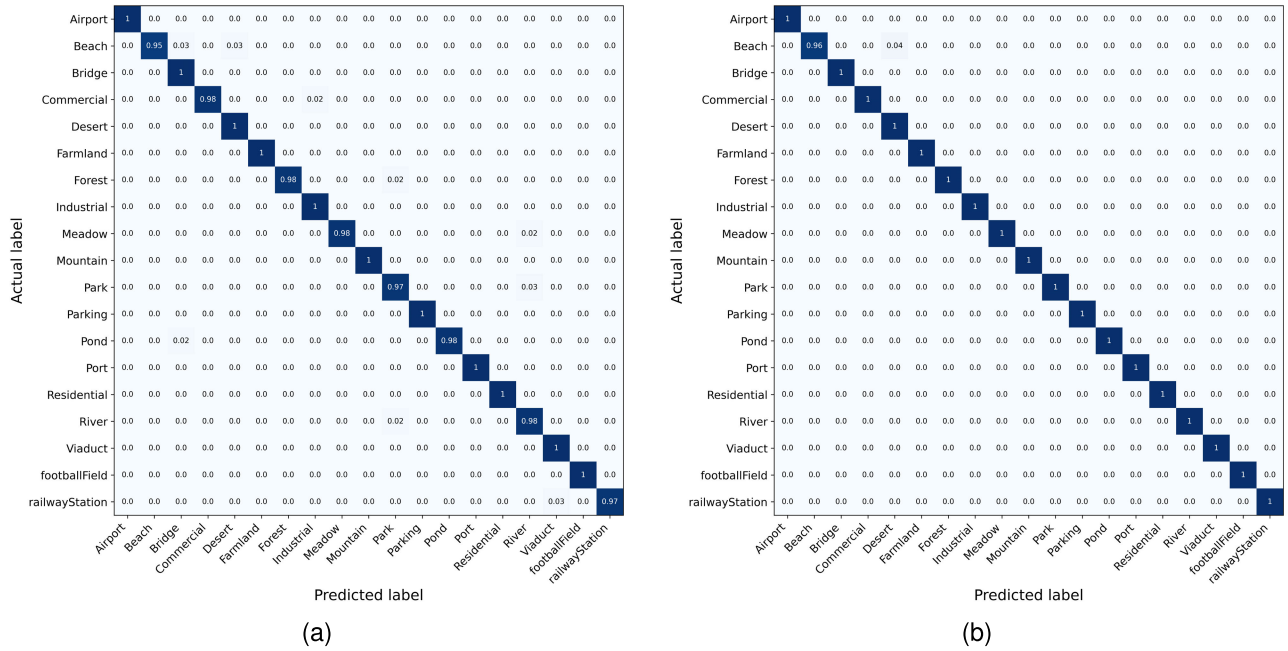


Fig. 4. Confusion matrices of DBEBiMSC under different training ratios on the WHU-RS19 dataset. (a) 20%. (b) 50%.

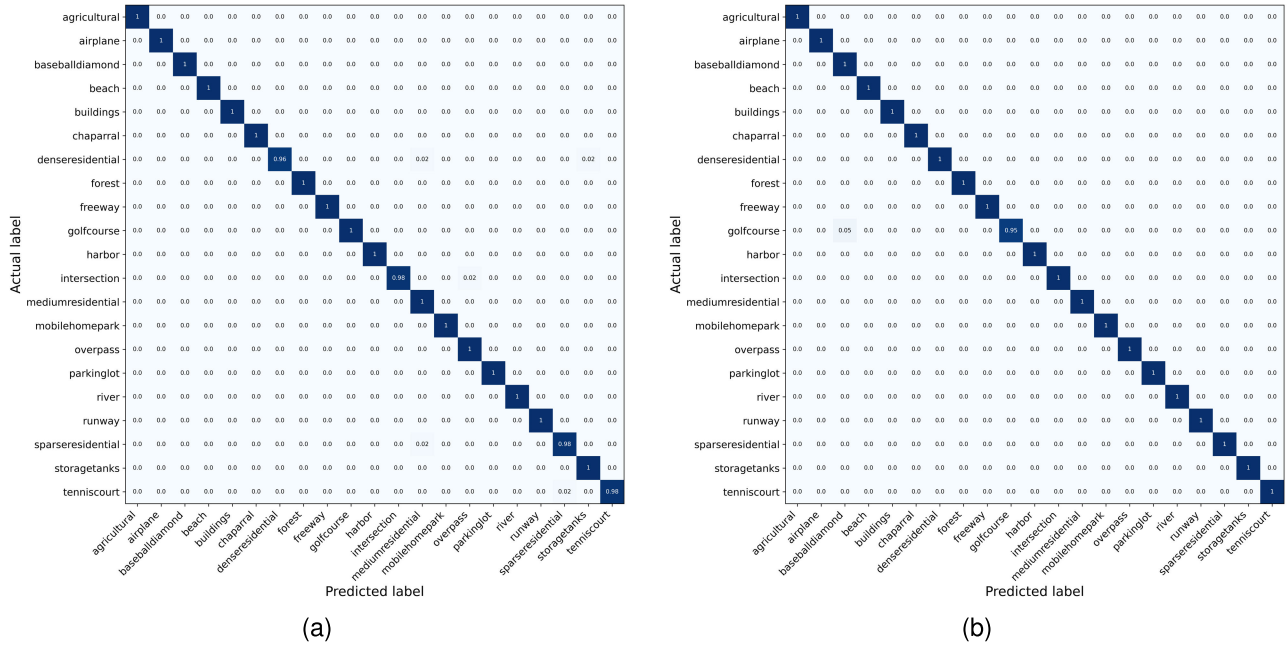
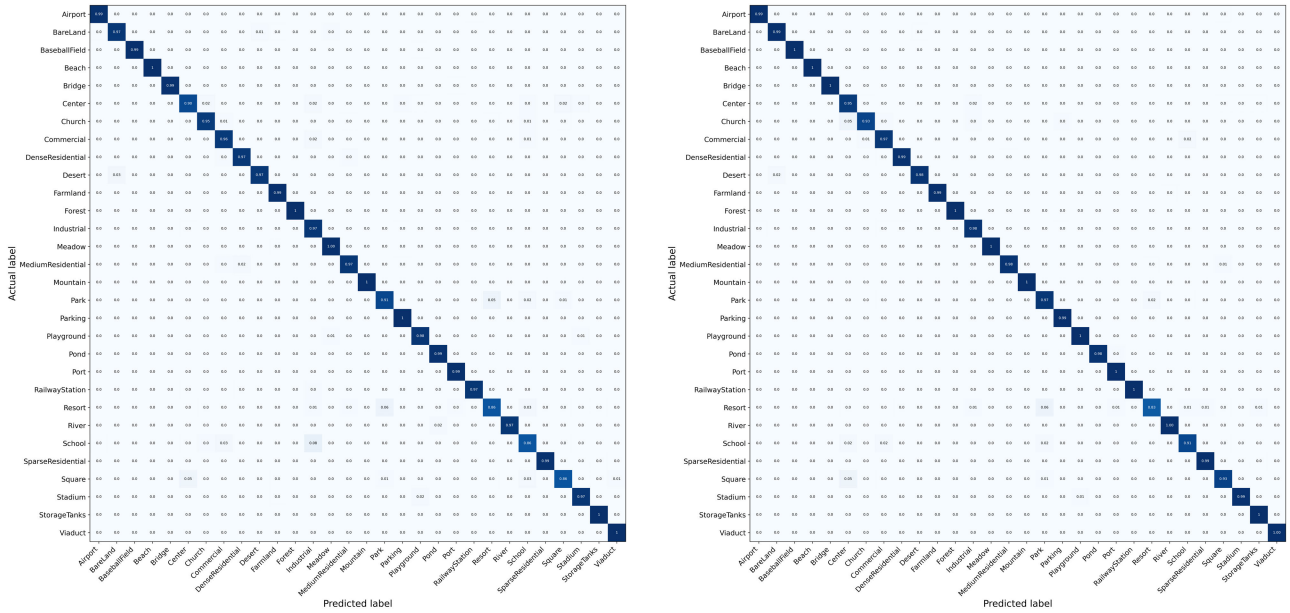


Fig. 5. Confusion matrices of DBEBiMSC under different training ratios on the UCM dataset. (a) 50%. (b) 80%.

diagonal, and the remaining errors are mainly caused by high visual similarity between a few closely related categories, with the most representative confusion occurring between *beach* and *desert*. When the training ratio increases to 50%, the matrix becomes nearly perfect and only sporadic residual confusion remains.

2) UCM Dataset: We evaluate DBEBiMSC on the UCM dataset against representative CNN, Transformer, and hybrid methods. As shown in Table II, DBEBiMSC achieves the best OA among all compared approaches, reaching 99.24% $\pm$ 0.16% and 99.86% $\pm$ 0.13% under the 50% and 80% training ratios, respectively. Compared with classical CNN baselines, our model yields consistently higher accuracy and stronger stability, it also slightly outperforms recent multiscale or feature fusion networks. Against strong hybrid architectures, DBEBiMSC delivers marginal yet consistent gains while preserving a compact model size, validating the effectiveness of the proposed cross-branch fusion. Fig. 5(a) and (b) shows the confusion matrices under the 50% and 80% training ratios. With 50% training data, predictions are already strongly diagonal and the remaining errors are mainly caused by high visual similarity within a few closely related scene groups, including the residential density continuum (e.g., *dense residential*, *medium residential*,

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(a) (b)

Fig. 6. Confusion matrices of DBEBiMSC under different training ratios on the AID dataset. (a) 20%. (b) 50%.

TABLE II

COMPARISON OF OA AND STD OF DIFFERENT METHODS ON UCM. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Year	OA±STD(%)	
		50% Training	80% Training
(† : CNN-based methods)			
GoogLeNet [39]†	2017	92.70±0.60	94.31±0.89
VGGNet16 [39]†	2017	94.14±0.69	95.21±1.20
ResNet50 [33]†	2016	98.30±0.20	98.95±0.80
DCNNet [28]†	2023	95.21±0.31	98.87±0.26
CSCANet [41]†	2023	-	99.76±0.16
MGSNet [42]†	2023	-	99.76±0.14
MobileNetV4-Large [23]†	2024	94.29±0.86	98.00±0.90
CGINet [27]†	2024	-	99.84±0.16
GLR-CNN [2]†	2024	97.52±0.33	99.14±0.26
(‡ : ViT-based or Hybrid architecture-based methods)			
ViT-B [33]‡	2020	97.30±0.32	98.81±0.43
SCViT [22]‡	2022	98.90±0.19	99.57±0.31
ET-GSNet [43]‡	2022	-	99.29±0.34
EMTCAL [34]‡	2022	98.67±0.16	99.57±0.28
LDBST [29]‡	2023	98.76±0.38	99.52±0.24
MHCT [35]‡	2025	98.97±0.11	99.84±0.16
CGKD [44]‡	2025	97.14	98.57
ACTFormer [32]‡	2025	99.14±0.09	99.76±0.26
LSDGNet [33]‡	2025	99.03±0.07	99.57±0.27
STMSF [36]‡	2025	99.01±0.31	99.58±0.23
Ours†	2025	**99.24±0.16**	**99.86±0.13**

TABLE III

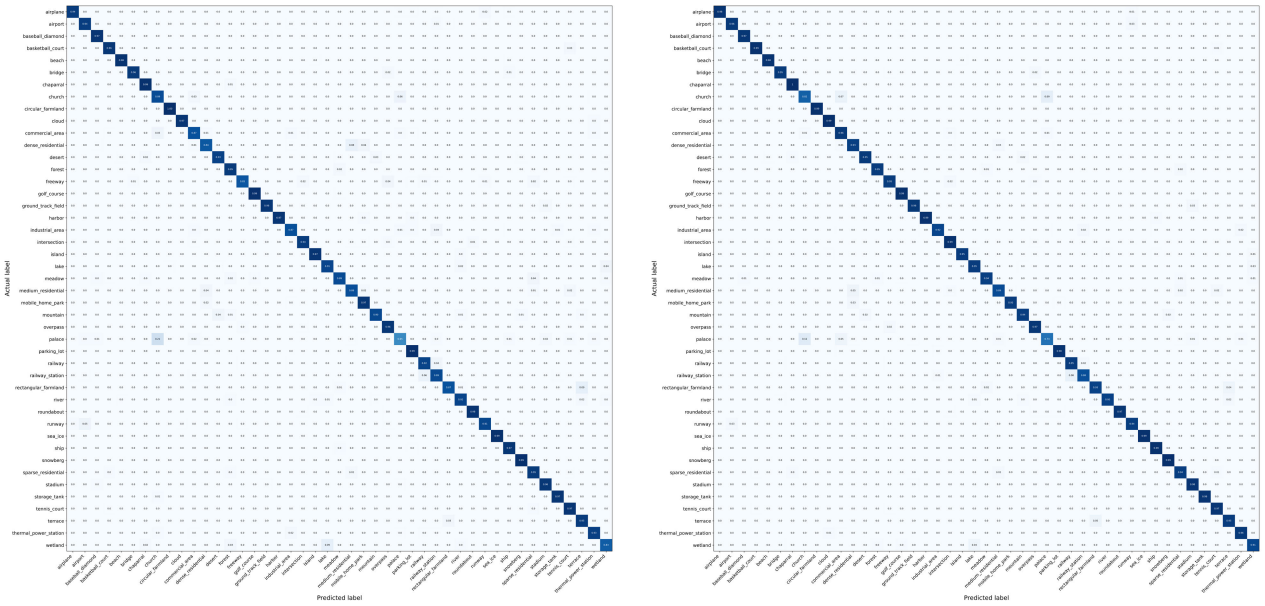
COMPARISON OF OA AND STD OF DIFFERENT METHODS ON AID. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Year	OA±STD(%)	
		20% Training	50% Training
(† : CNN-based methods)			
GoogLeNet [39]†	2017	83.44±0.40	86.39±0.55
VGGNet16 [39]†	2017	86.59±0.29	89.64±0.36
ResNet50 [33]†	2016	92.57±0.17	95.96±0.17
DCNNet [28]†	2023	93.64±0.25	96.23±0.19
MobileNetV4-Large [23]†	2024	93.05±0.13	94.76±0.58
CSCANet [41]†	2023	94.67±0.20	96.83±0.14
MGSNet [42]†	2023	95.46±0.21	97.18±0.16
CGINet [27]†	2024	95.35±0.14	97.10±0.24
GLR-CNN [2]†	2024	93.09±0.16	95.64±0.11
MSCN [3]†	2025	95.86±0.16	97.46±0.12
(‡ : ViT-based or Hybrid architecture-based methods)			
ViT-B [33]‡	2020	94.80±0.46	96.66±0.46
SCViT [22]‡	2022	95.56±0.17	96.98±0.16
ET-GSNet [43]‡	2022	95.58±0.18	96.88±0.19
EMTCAL [34]‡	2022	94.69±0.14	96.41±0.23
LDBST [29]‡	2023	95.10±0.09	96.84±0.20
CGKD [44]‡	2025	92.31	95.42
LSDGNet [33]‡	2025	95.81±0.14	97.49±0.19
STMSF [36]‡	2025	96.15±0.16	97.51±0.37
Ours†	2025	**96.26±0.29**	**97.79±0.22**

and *sparse residential*) and layout-level ambiguity in road-junction categories (e.g., *intersection* versus *overpass*). When the training ratio increases to 80%, the matrix becomes nearly perfect and only sporadic residual confusion remains.

3) AID Dataset: We further evaluate DBEBiMSC on the large-scale AID benchmark to assess its performance under complex multiclass settings. As shown in Table III, DBEBiMSC obtains 96.26% ± 0.29% and 97.79% ± 0.22% OA under 20% and 50% training ratios, ranking first among all competitors. It delivers clear gains over CNN baselines and slightly surpasses recent ViT/hybrid methods such as STMSF and LSDGNet while remaining compact, indicating strong generalization on this challenging dataset. Fig. 6(a) and (b) shows the confusion matrices under the 20% and 50% training ratios. With 20% training data, the matrix is already strongly diagonal, and the remaining errors are mainly concentrated in a few highly correlated fine-grained categories. These residual confusions primarily arise from texture-similar land-cover pairs (e.g., *bareland* versus *desert*), semantic overlap among mixed-use built-up scenes (e.g., *commercial* versus *industrial*), and layout-level similarity among landmark-like categories (e.g., *center* versus *square*). When the training ratio increases to 50%, these confusions are markedly reduced, and

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(a) (b)

Fig. 7. Confusion matrices of DBEBiMSC under different training ratios on the NWPU-RESISC45 dataset. (a) 10%. (b) 20%.

TABLE IV

COMPARISON OF OA AND STD OF DIFFERENT METHODS ON NWPU. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Method	Year	OA±STD(%)	
		10% Training	20% Training
(† : CNN-based methods)			
GoogLeNet [33]†	2017	76.19±0.38	78.48±0.26
VGGNet16 [33]†	2017	76.47±0.18	79.79±0.15
ResNet50 [33]†	2016	88.48±0.21	91.86±0.19
CSCANet [41]†	2023	91.27±0.11	93.72±0.10
MGSNet [42]†	2023	92.40±0.16	94.57±0.12
CGINet [27]†	2024	92.28±0.17	94.38±0.13
GLR-CNN [2]†	2024	89.35±0.25	92.11±0.22
MSCN [3]†	2025	92.64±0.09	94.59±0.11
(‡ : ViT-based or Hybrid architecture-based methods)			
ViT-B [33]‡	2020	91.63±0.63	93.97±0.14
SCViT [22]‡	2022	92.72±0.04	94.66±0.10
ET-GSNet [43]‡	2022	92.72±0.28	94.50±0.18
EMTCAL [34]‡	2022	91.63±0.19	93.65±0.12
MHCT [35]‡	2025	92.76±0.13	93.67±0.29
CGKD [44]‡	2025	89.17	92.35
ACTFormer [32]‡	2025	92.85±0.14	94.76±0.21
STMSF [36]‡	2025	92.88±0.16	94.95±0.11
Ours†	2025	**92.95±0.14**	**94.99±0.14**

only a small number of sporadic residual errors remain, still mainly within the same challenging clusters.

4) NWPU Dataset: Experimental results on the NWPU dataset further validate the effectiveness of DBEBiMSC for large-scale, multiclass RSSC. As reported in Table IV, DBEBiMSC achieves OAs of 92.95% ± 0.14% and 94.99% ± 0.14% under the 10% and 20% training ratios, respectively, outperforming all competing methods. These improvements remain consistent when compared with recent ViT-based and hybrid architectures while preserving a compact architecture with convolutional backbones. Fig. 7(a) and (b) shows the confusion matrices under the 10% and 20% training ratios. With only 10% training data, DBEBiMSC already produces a clearly diagonal confusion matrix, and the remaining errors are mainly concentrated in several highly correlated fine-grained scene groups. These residual confusions are primarily associated with shared structural patterns in aviation-related categories (e.g., *airport* versus *runway*), layout-level similarity among landmark-like buildings (e.g., *palace* versus *church*), and water-dominated scenes with weak boundary cues (e.g., *wetland* versus *lake*). When the training ratio increases to 20%, these ambiguities are further reduced and the diagonal dominance becomes more pronounced, with only a few sporadic residual confusions remaining within the same challenging groups. This result further confirms the improved interclass separability of DBEBiMSC on NWPU.

To further understand the residual errors, we summarize three recurrent failure modes observed across the benchmarks. First, texture-close and boundary-ambiguous land-cover scenes remain challenging when discriminative cues are weak, leading to occasional confusions between visually similar categories such as *beach* and *desert*, *bareland* and *desert*, and *wetland* and *lake*. Second, semantic overlap in built-up scenes yields errors among functionally related categories with similar layouts, where *commercial* and *industrial* areas as well as residential density subclasses are prone to be confused. Third, fine-grained layout similarity also causes misclassification between classes sharing highly correlated geometric structures, such as *airport* versus *runway* and *palace* versus *church*. These observations suggest that future improvements may require more boundary-sensitive and region-aware modeling to better preserve subtle discriminative cues in highly similar or mixed scenes.

D. Parameter Sensitivity Analysis

To assess the robustness of the proposed sparsification to the Top-k setting, we conduct a controlled sweep over $(k_1, k_2) \in \{(1,1),(2,3),(2,4),(2,6),(4,3),(4,6),(8,6)\}$ under

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TABLE V

TOP-k SENSITIVITY ANALYSIS OF EBIMSC UNDER DIFFERENT (k_1, k_2) SETTINGS. ALL VALUES ARE OA (%). THE BEST RESULTS IN EACH COLUMN ARE HIGHLIGHTED IN BOLD

(k_1, k_2)	WHU-RS19		UCM		AID		NWPU	
	20% Training	50% Training	50% Training	80% Training	20% Training	50% Training	10% Training	20% Training
$(1, 1)$	98.12	99.60	99.14	**100.00**	96.26	97.68	93.10	94.99
$(2, 3)$	98.00	99.60	99.52	99.76	96.20	97.78	92.96	**95.08**
$(2, 4)$	**98.37**	99.60	99.24	**100.00**	**96.41**	97.74	**93.23**	94.78
$(2, 6)$	98.25	99.60	99.33	**100.00**	96.25	97.94	92.99	94.91
$(4, 3)$	**98.37**	**100.00**	**99.62**	**100.00**	96.33	97.88	92.99	94.96
$(4, 6)$	98.12	99.40	99.43	99.76	96.16	97.76	92.85	94.75
$(8, 6)$	98.12	99.60	99.43	**100.00**	96.09	**97.96**	93.02	94.85

TABLE VI

ABLATION RESULTS OF EACH MODULE CONFIGURATION IN THE PROPOSED METHOD

Backbone	AES	EBiMSC			UCM(50%)	AID(50%)
		BiMSC	Enhanced Path			
✓					98.71 ± 0.16	97.58 ± 0.29
✓	✓				98.74 ± 0.21	97.59 ± 0.18
✓	✓	✓			99.12 ± 0.27	97.77 ± 0.29
✓	✓	✓		✓	$\mathbf{99.24 \pm 0.16}$	$\mathbf{97.79 \pm 0.22}$

TABLE VII

COMPARISON OF FUSION STRATEGIES UNDER THE SAME DUAL-BACKBONE ARCHITECTURE AND TRAINING PROTOCOL. NOTE: THE BACKBONE+AES CONFIGURATION IS KEPT FIXED, AND ONLY THE FUSION MODULE IS REPLACED. CONCAT IS USED AS THE DEFAULT FUSION BASELINE

Fusion	Params (M)	GFLOPs	UCM(50%)	AID(50%)
Concat	19.40	4.236	98.74 ± 0.21	97.59 ± 0.18
MSC	19.25	4.230	98.63 ± 0.29	97.14 ± 0.28
AFF	19.20	4.227	98.80 ± 0.13	97.59 ± 0.32
BiCrossAttn	19.25	4.233	98.93 ± 0.17	97.62 ± 0.21
BiMSC	19.32	4.233	99.12 ± 0.27	97.77 ± 0.29
EBiMSC	19.50	4.242	$\mathbf{99.24 \pm 0.16}$	$\mathbf{97.79 \pm 0.22}$

the same dual-backbone architecture and training protocol. We report single-run OA across four datasets and multiple training ratios in Table V. Overall, performance remains stable over a broad sparsity spectrum, with OA variations typically below 0.4% and the maximum variation not exceeding 0.60% across all dataset/ratio settings. For example, the variation is within 0.48% on UCM (50% training) and within 0.28% on AID (50% training). These results indicate that EBiMSC is not overly sensitive to a particular Top-k choice and that retaining only a fraction of the strongest attention responses does not cause severe semantic degradation in practice. We emphasize that this sweep is provided as a robustness study rather than dataset-specific hyperparameter tuning. Unless otherwise specified, we adopt a unified default $(k_1, k_2) = (2, 3)$ for all datasets without dataset-specific tuning. Moreover, EBiMSC introduces a parallel enhanced fusion path without Top-k pruning, serving as a dense refinement branch to preserve potentially low-response yet semantically important cues (e.g., small objects and weak-texture categories) that may be underemphasized in the sparse attention route.

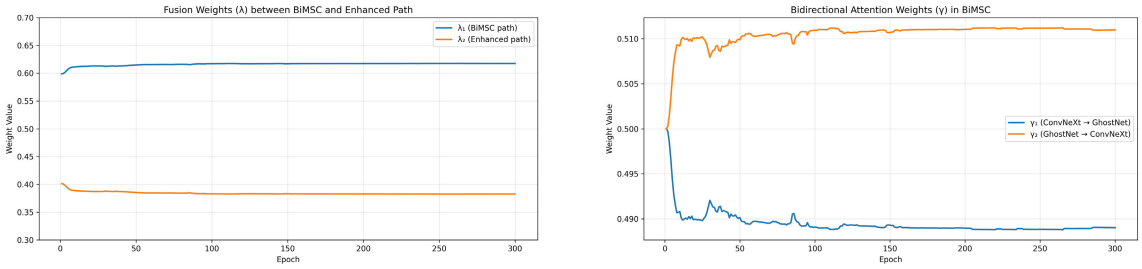
E. Ablation Studies

This section reports ablation experiments to verify the effectiveness and design rationale of each component in DBEBiMSC. All ablations are conducted on UCM and AID under identical training settings. We progressively add or replace modules to isolate their contributions to classification performance.

1) Effectiveness of Each Module: Table VI summarizes the results of different module configurations. Incrementally adding attention enhancement scheme (AES, composed of triplet attention and GCE), BiMSC, and the enhanced path consistently improves accuracy on both datasets. The baseline achieves 98.71% OA on UCM and 97.58% on AID. Adding AES yields a small yet stable gain to 98.74% and 97.59%. Introducing BiMSC raises OA to 99.12% and 97.77%, confirming the effectiveness of bidirectional multi-scale cross-branch interaction. Enabling the enhanced path further improves performance to 99.24% and 97.79%, i.e., +0.53% and +0.21% over the baseline on UCM and AID. Given the negligible complexity increase, these modules effectively enhance representation and discrimination under a lightweight budget.

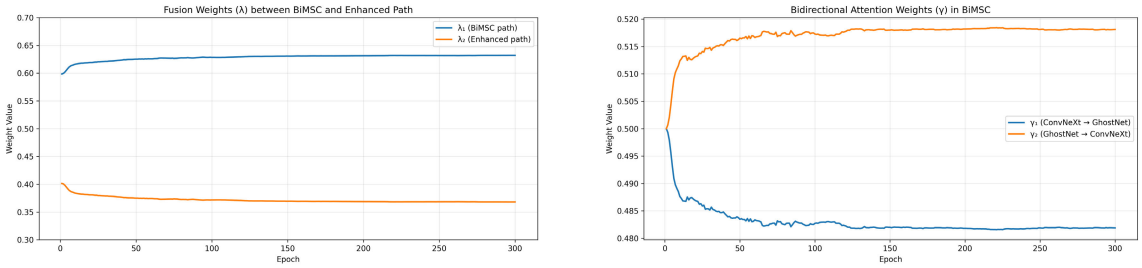
2) Comparison of Fusion Strategies: To validate the necessity of EBiMSC, we conduct controlled comparisons under the same dual-backbone architecture and training protocol, where only the fusion module is varied while all other components remain fixed. As shown in Table VII, replacing the default Concat fusion with the original single-direction MSC in our heterogeneous dual-branch network leads to lower OA than the Backbone+AES+Concat baseline. Specifically, OA decreases from 98.74% to 98.63% on UCM and from 97.59% to 97.14% on AID (50% training). This indicates that the one-way MSC, originally designed for intra-backbone hierarchical fusion, is not suitable for cross-branch semantic–detail cooperation and can introduce biased information flow between heterogeneous representations. In addition, we compare BiMSC/EBiMSC with representative fusion baselines of approximate complexity, including AFF (an attention-based gating mechanism) and BiCrossAttn (a lightweight symmetric bidirectional cross-attention module). These variants are implemented by replacing only the fusion operator and operate under approximate parameter/FLOP budgets to BiMSC/EBiMSC. Specifically, AFF and BiCrossAttn achieve 98.80% and 98.93% on UCM, and 97.59% and 97.62% on AID, respectively. Notably, BiCrossAttn serves as a controlled baseline to isolate the effect of bidirectional

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(a) (b)

Fig. 8. Training dynamics of adaptive fusion and attention weights in the proposed EBiMSC on the UCM dataset. The curves show smooth convergence and consistent adaptation, indicating stable weighting between the BiMSC and enhanced fusion paths as well as balanced bidirectional attention. (a) UCM: fusion weights (λ_1 and λ_2). (b) UCM: attention coefficients (γ_1 and γ_2).



(a) (b)

Fig. 9. Training dynamics of adaptive fusion and attention weights in the proposed EBiMSC on the AID dataset. The curves exhibit stable convergence, revealing consistent learning behavior with stable weighting between the two fusion paths and balanced bidirectional attention across different RS scenes. (a) AID: fusion weights (λ_1 and λ_2). (b) AID: attention coefficients (γ_1 and γ_2).

TABLE VIII

COMPARISON OF PARAMETERS AND GFLOPS OF THE PROPOSED MODEL UNDER DIFFERENT MODULE CONFIGURATIONS

Backbone	AES	EBiMSC			Params (M)	GFLOPs
		BiMSC	Enhanced Path			
✓					19.26	4.229
✓	✓				19.40	4.236
✓	✓	✓			19.32	4.233
✓	✓	✓		✓	19.50	4.242

interaction before introducing the multiscale sparse design and the enhanced compensation path in BiMSC/EBiMSC. In contrast, BiMSC improves OA to 99.12% on UCM and 97.77% on AID, suggesting that bidirectional cross-branch interaction plays an important role in this heterogeneous setting. Building upon BiMSC, EBiMSC further improves OA to **99.24**% on UCM and **97.79**% on AID with only marginal overhead, supporting the effectiveness of the enhanced compensation path and adaptive weighting. Together with the ablation results in Table VI and the complexity analysis in Table VIII, these controlled comparisons provide strong evidence that EBiMSC provides a necessary redesign tailored for stable bidirectional cross-branch fusion in our dual-branch CNN.

3) Discussion: The ablations confirm that AES slightly boosts fine-grained feature extraction, BiMSC brings the main gains via bidirectional cross-scale fusion, and the enhanced path provides additional improvement at negligible cost. Additionally, comparisons with fusion baselines such as AFF and BiCrossAttn show that our bidirectional designs, particularly EBiMSC, achieve more competitive performance under approximate computational budgets. The stepwise comparisons reflect a progressive refinement toward adaptive and efficient cross-branch interaction within a CNN-based framework.

F. Training Dynamics and Interpretability

To better understand the behavior of EBiMSC, we track the evolution of the adaptive fusion weights (λ_1 and λ_2) and the bidirectional attention coefficients (γ_1 and γ_2) during training on the UCM and AID datasets. Here, λ_1 and λ_2 quantify the relative contributions of the main BiMSC stream and the enhanced fusion path, while γ_1 and γ_2 regulate the two opposite-direction multiscale interactions within BiMSC. As shown in Figs. 8 and 9, all coefficients evolve smoothly and converge steadily without noticeable oscillation, indicating stable optimization throughout training. The fusion weights quickly settle into a consistent allocation in which the main BiMSC path is slightly emphasized and the enhanced path remains a complementary contributor, validating its role as a local-detail compensator. Meanwhile, the bidirectional coefficients converge to a near-balanced state on both datasets, suggesting that cross-scale information is jointly exploited in two directions rather than dominated by a single route. In particular, the trajectories indicate that both directions and both fusion paths remain active throughout training, rather than collapsing to a single dominant stream, which supports their complementary roles.

Overall, these dynamics demonstrate that EBiMSC learns an interpretable, data-driven balance between global semantics

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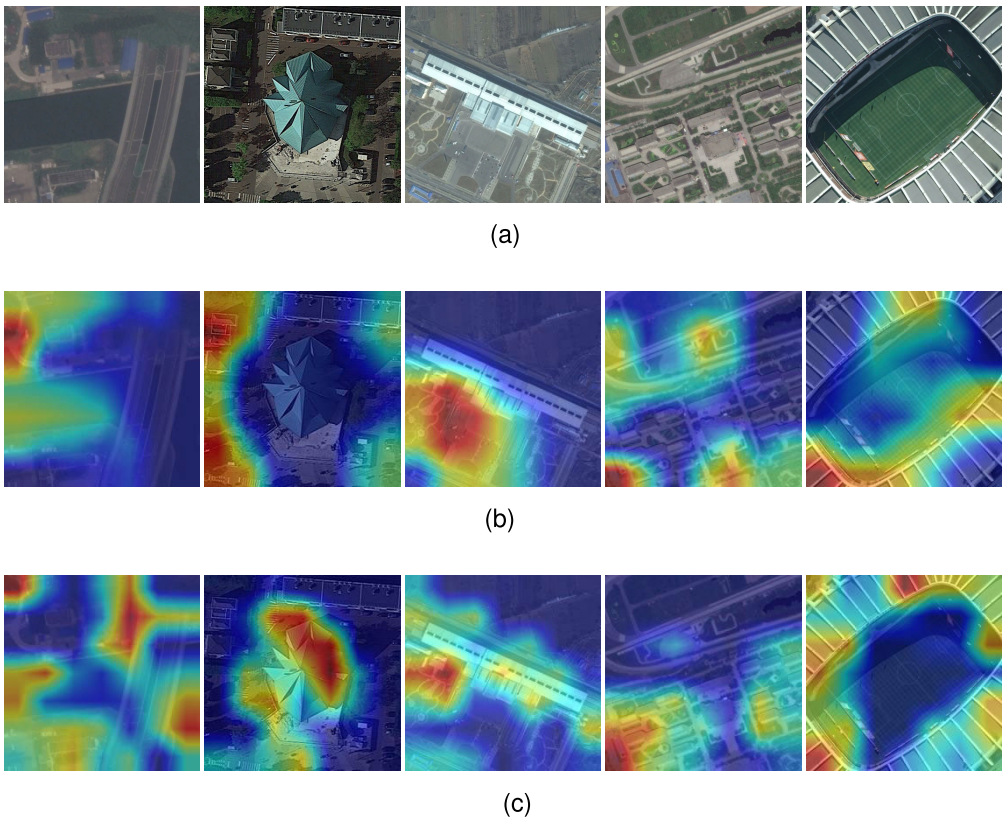


Fig. 10. Visualization comparison of classification results with and without the EBiMSC module. The first row shows the original input images, the second row displays the results without EBiMSC, and the third row presents the results with EBiMSC. The visualizations indicate that incorporating the EBiMSC module leads to more concentrated and semantically accurate attention regions across all five scene categories: bridge, center, railway station, school, and stadium. (a) Original images. (b) Without EBiMSC. (c) With EBiMSC.

and fine-grained details, without relying on manually fixed fusion ratios or handcrafted routing rules.

G. Parameter Quantity and Computational Complexity

Table VIII reports the parameters and GFLOPs of DBEBiMSC under different module configurations. The baseline dual-branch backbone has 19.26 M parameters and 4.229 GFLOPs. Adding the AES incurs only a negligible overhead (19.40 M/4.236 GFLOPs). Further introducing BiMSC maintains a similar complexity level (19.32 M/4.233 GFLOPs), indicating that bidirectional sparse fusion is lightweight. With the enhanced fusion path enabled, the full model reaches 19.50 M parameters and 4.242 GFLOPs, i.e., only about 1.2% more parameters and 0.3% more FLOPs than the baseline, while achieving the best ablation accuracy. These results confirm that the proposed modules improve performance without materially increasing computational cost.

Table IX compares DBEBiMSC with representative Transformer-based, hybrid, and CNN-based methods. Transformer and hybrid architectures, including SCViT, MHCT, SF-MSFormer, LGLFormer, and ACTFormer, typically incur higher computational cost due to tokenization and self-attention operations. MSCN, although CNN-based, still adopts a relatively heavy configuration in both parameters and FLOPs. In contrast, DBEBiMSC preserves a heterogeneous dual-branch design with purely convolutional backbones, achieving substantially fewer parameters and lower GFLOPs while maintaining SOTA accuracy. This indicates a more favorable accuracy–efficiency tradeoff than recent Transformer/hybrid models and heavier CNN counterparts under practical resource budgets. Moreover, under single-image inference with 224 × 224 inputs and a batch size of 1, DBEBiMSC reaches an average latency of **17.8 ms per image** (about 56 FPS), which aligns well with the theoretical complexity advantages in Tables VIII and IX.

TABLE IX
PARAMETERS AND GFLOPS COMPARISON WITH REPRESENTATIVE TRANSFORMER-, HYBRID-, AND CNN-BASED METHODS COMMONLY REPORTED

Method	Year	Params (M)	GFLOPs (G)
SCViT	2022	40.1	9.3
SF-MSFormer	2023	26.12	19.56
LGLFormer	2024	22.96	14.29
MHCT	2025	22.92	15.78
ACTFormer	2025	23.04	-
MSCN	2025	38.37	6.42
Ours	2025	**19.50**	**4.24**

H. Visualization

To further validate the explainability and feature extraction capabilities of the proposed DBEBiMSC model, we used Grad-CAM [45] to visualize the attention maps of the model on the AID dataset. Fig. 10 illustrates the feature visualization

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results for five representative scene categories: bridge, center, railway station, school, and stadium. As shown in the second row of Fig. 10, the baseline model without EBiMSC yields attention maps that are relatively diffuse and scene-dependent. In the bridge example, activations drift to surrounding background and outer edges rather than the bridge body itself; in the center scene, the model responds to broad urban textures instead of the true core region. After introducing EBiMSC (third row), the attention becomes noticeably more concentrated and better aligned with the discriminative objects: the bridge and urban center are sharply highlighted with reduced background leakage. Similar improvements are observed for the railway station, school, and stadium, where the model focuses on the station building, classroom blocks, and stadium field/outline instead of nearby clutter.

These qualitative results indicate that EBiMSC strengthens cross-branch feature interaction and suppresses redundant responses, guiding the network toward semantically meaningful regions. By promoting bidirectional multiscale aggregation between global semantics and local details, the module enhances object-aligned localization and improves separability for visually complex RS scenes.

V. Conclusion

This article proposes DBEBiMSC, a dual-branch convolutional network for RSSC. The model builds a heterogeneous dual-backbone architecture based on ConvNeXt-Tiny and GhostNetV3, and integrates the proposed EBiMSC module to jointly capture global semantic context and local fine-grained details in a CNN framework. In addition, two lightweight attention mechanisms, triplet attention and GCE, are incorporated at the branch and fusion stages to further enhance cross-branch feature interaction and fully exploit the representational potential of the dual-backbone design. Extensive experiments on four public RSSC datasets demonstrate that DBEBiMSC achieves consistently superior performance compared with recent CNN-based, Transformer-based, and hybrid methods, while maintaining a compact model size and low computational cost. These results indicate that the proposed architecture effectively mitigates the challenges of high interclass similarity and substantial intraclass diversity in RS imagery. Overall, this work not only reaffirms the potential of CNNs for RSSC but also provides a practical perspective for designing lightweight and efficient dual-branch CNNs with cross-scale feature fusion. In future work, exploiting RS-specific pretraining to boost representational capacity can be a promising direction. Nevertheless, DBEBiMSC may still be prone to misclassification for closely related scene categories with high visual similarity, and improving robustness under such failure modes will be a key focus.

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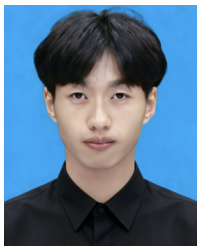
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